

Biomechanical Impact Performance Assessment of 3D-Printed and Plaster Short Arm Wrist Casts

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Background

Wrist fractures, particularly the distal radius fractures (DRF) (Figure 1), are prevalent skeletal injuries in both older and active younger populations [1].

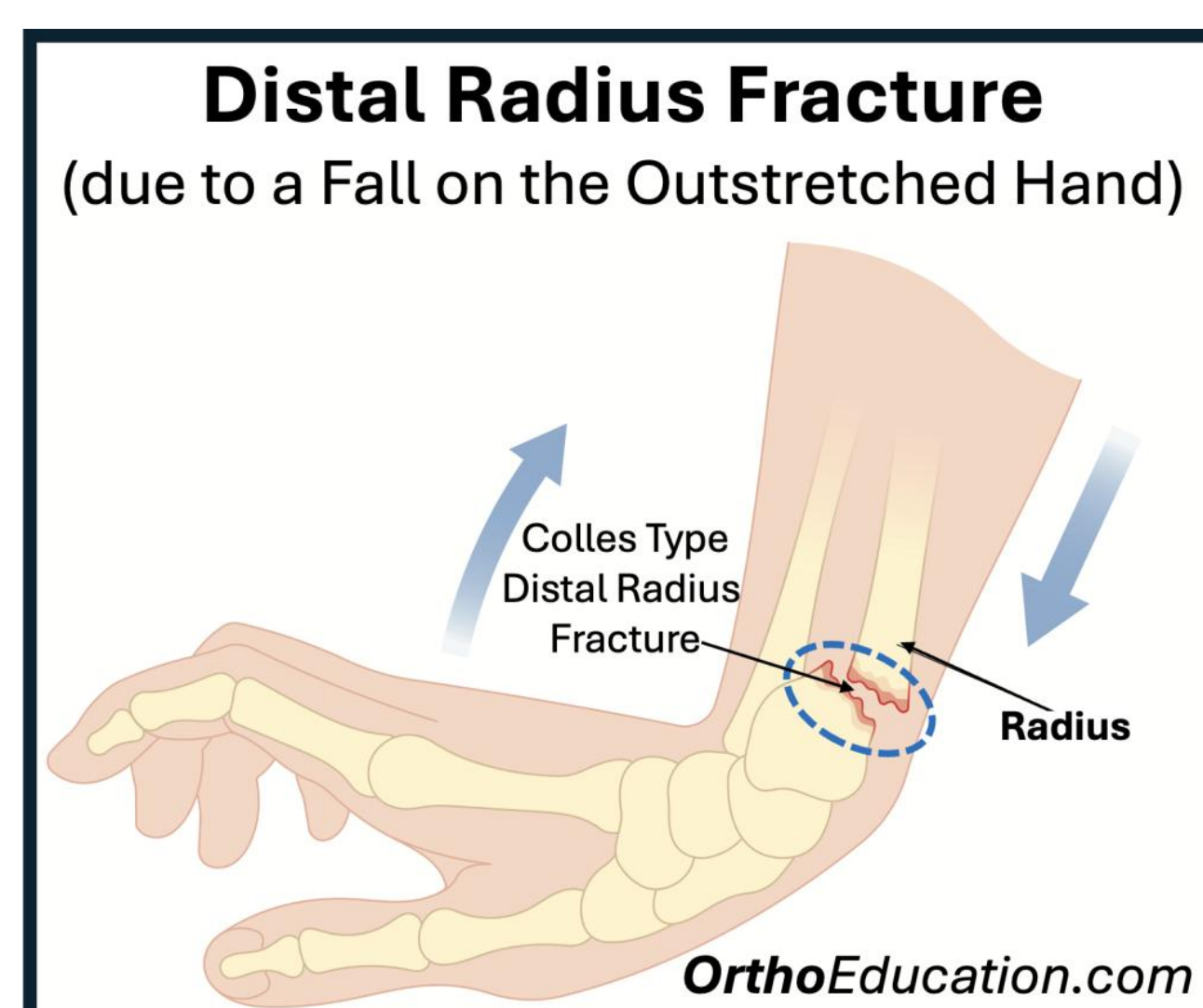


Figure 1. Injury Mechanism of DRF (modified from OrthoEducation.com)

DRF leads to huge losses of quality of life and great national expenditure for treatment and care (Figure 2) [1, 2].

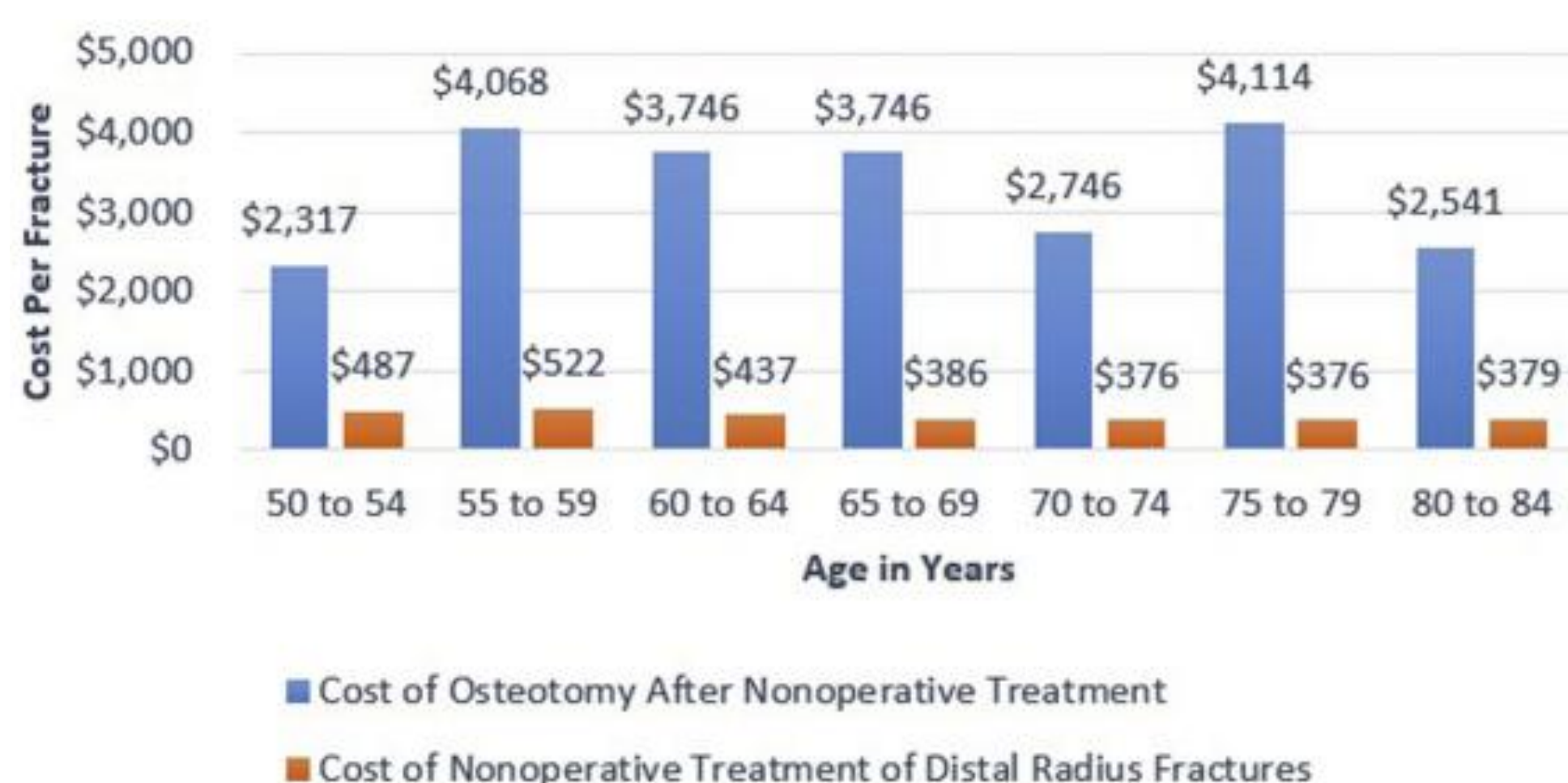


Figure 2. Relevant Treatment Costs of DRF [2]

National Health and Nutrition Examination Survey (NHANES) reports around 20% of patients suffering from **secondary or more** fractures after treatments.

Status Quo

Conventional Circumferential Plaster Casts (CPC):

- Gold standard non-operative immobilisation method
- Involving bandage & gypsum powder
- **Inconsistencies** in cast and padding quality and strict curing processes

Novel 3D-Printed Casts (3D cast) [3]:

- Higher predicted **customisability, comfort, convenience, and hygiene**
- Lower predicted **variability and cost** in manufacturing
- **Similar immobilization** ability to CPC
- **Protective performance under secondary impacts unknown**

Methodology

- A viscoelastic fingerless forearm model was 3D printed using the 95A thermoplastic polyurethane (TPU) (Figure 3).
- The 3D cast was 3D printed using polylactic acid (PLA) with a Voronoi cell structure (Figure 3).

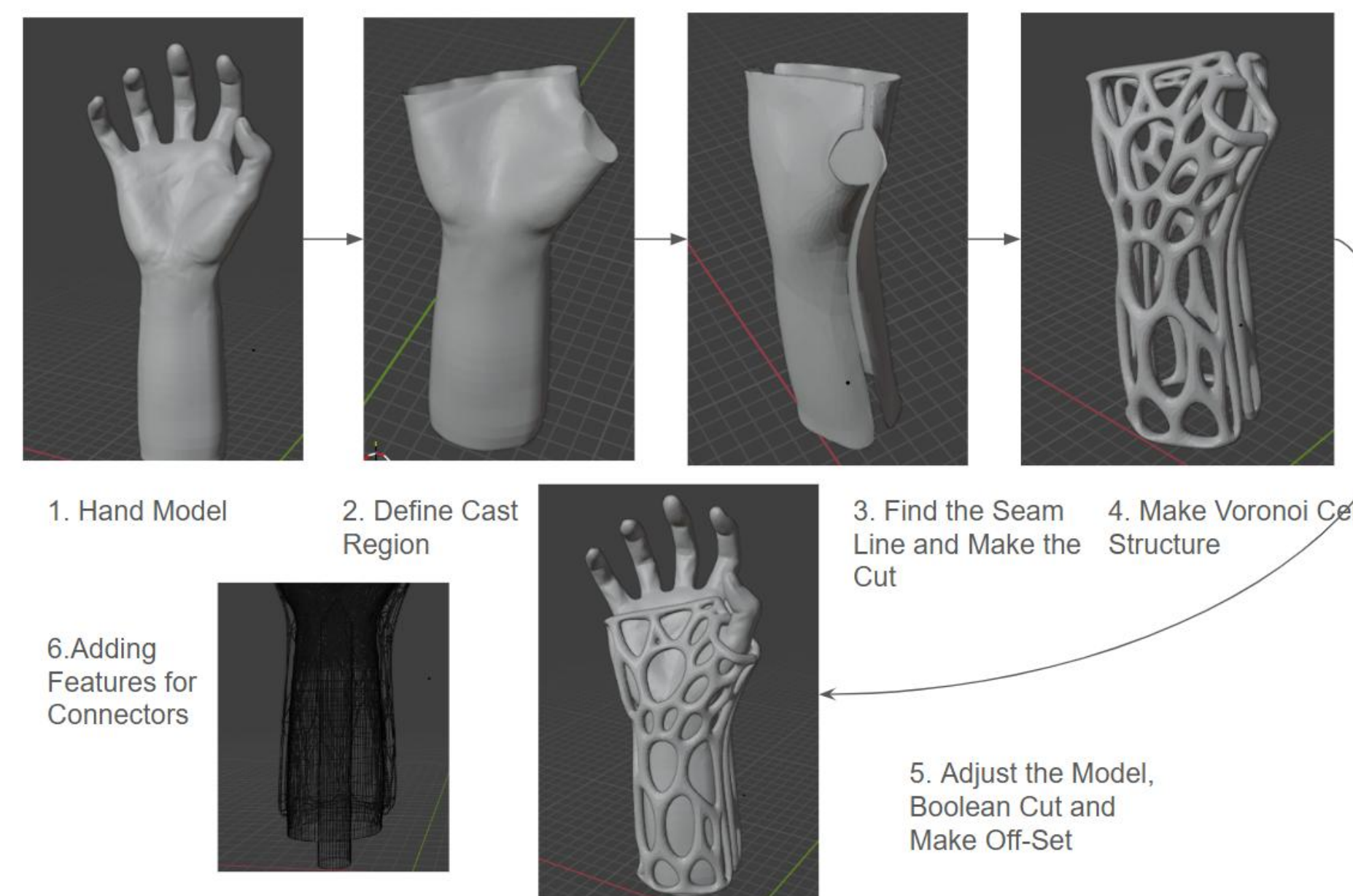


Figure 3. The Development of the forearm model and 3D cast

- The CPC was manufactured as per the standard medical procedures (Figure 4).



Figure 4. The Creation of the CPC

- The assembly of the forearm and cast was attached to a drop tower, mimicking a drop from 40 cm to a 50-degree surface, recorded by a high-speed camera and a 6-axis load cell (Figure 5).

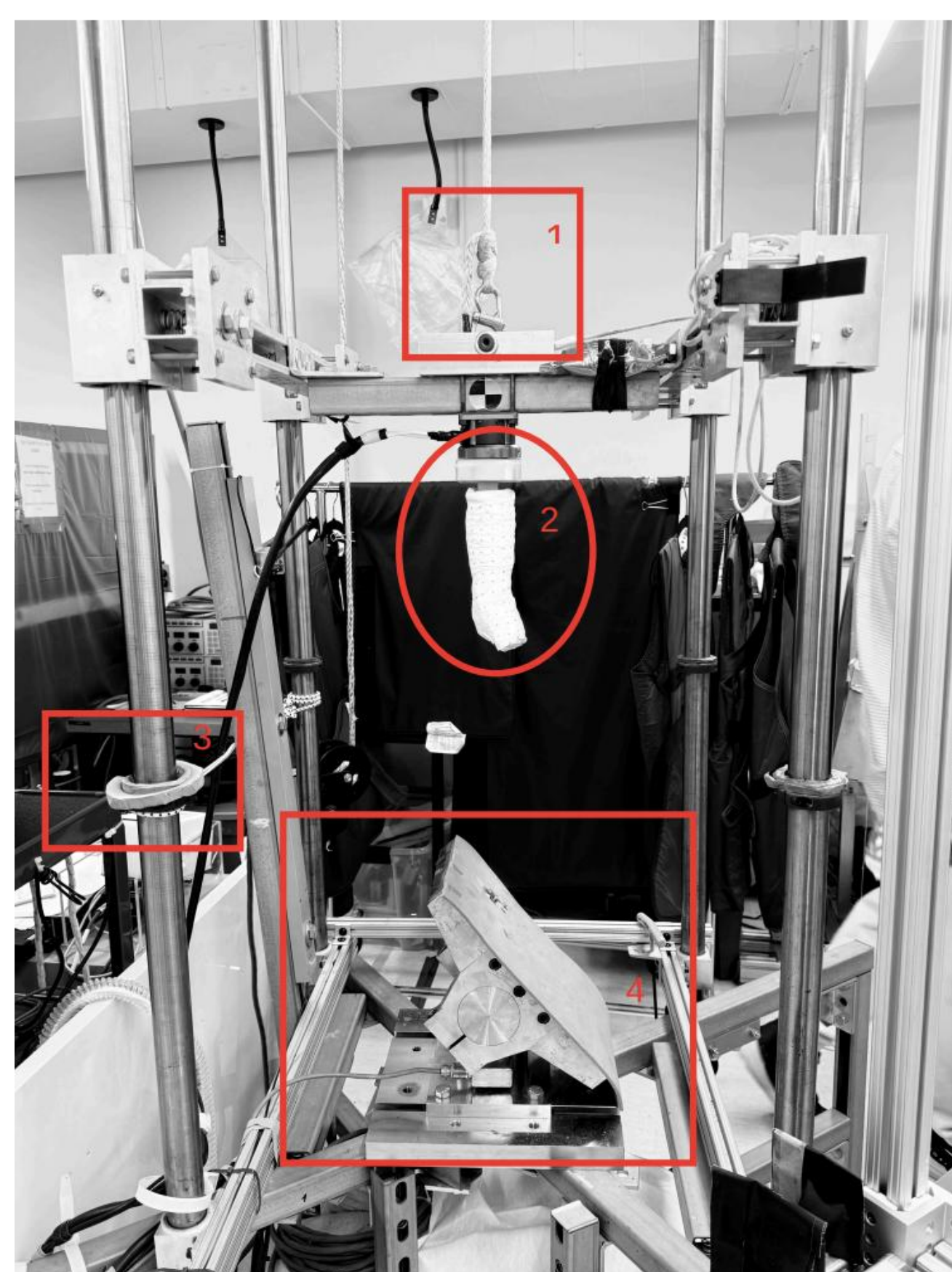


Figure 5. The Testing Setup

Results

- The 3D cast showed deflection and deformation in both vertical & horizontal directions and in the wrist angle (Figure 6).
- Peak axial force at 706 N and peak moment at 17 Nm (Table 1). Minor scratches shown.

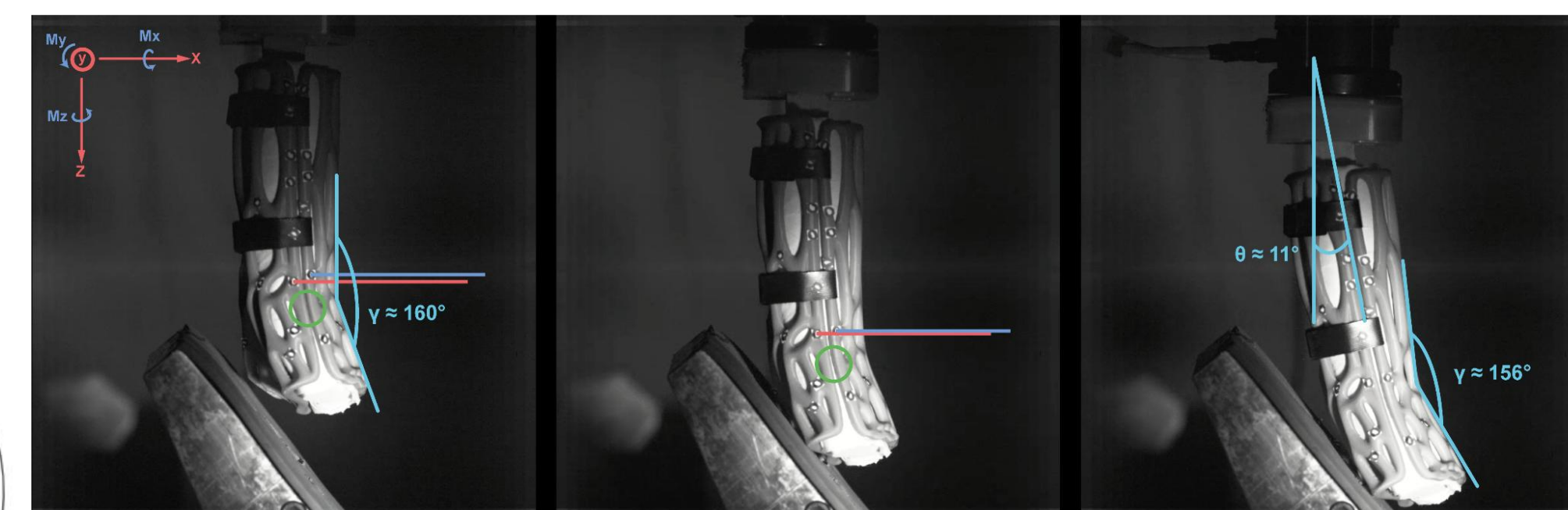


Figure 6. Footage of the 3D Cast Test

- The CPC exhibited similar deflection and deformation upon impact, mostly to a lesser extent (Figure 7).
- Peak axial force at 1049 N and peak moment at 9 Nm (Table 1). Cracks shown.



Figure 7. Footage of the CPC Test

Table 1. Peak Impact Forces and Moments

	3D Cast	CPC
Fx (N)	242.32	224.92
Fy (N)	-358.10	268.89
Fz (N)	-706.12	-1049.19
Mx (Nm)	12.95	6.41
My (Nm)	16.96	8.64
Mz (Nm)	1.76	-1.45

Conclusions

- A repeatable comparative experimental protocol for evaluating the protection provided by different cast designs
- Similar mechanical responses, with the 3D cast offering superior structural integrity.
- A foundation for future investigations of the 3D cast with more design variations, repetition, and evaluation metrics.

Acknowledgement

I would like to thank the Orthopaedic and Injury Biomechanics Group (OIBG) at the University of British Columbia (UBC) for lab and equipment access.